

IBONG TIRIRIT (PIPE 7)

Is the most common dryer used which consist of rotating cylinder inside which the materials flow while getting in contact with hot gas.

- A. Tower dryer
- B. Centrifugal dryer
- C. Trey dryer
- D. Rotary dryer

Ans: D

Is the ratio of the mass of water-vapor in air and the mass of air if it is saturated is called:

- A. Humidity ratio
- B. Mass ratio
- C. Vapor ratio
- D. Relative humidity

Ans: D

The hands feel painfully cold when the skin temperature reaches

- A. 8 deg C
- B. 10 deg C
- C. 12 deg C
- D. 14 deg C

Ans: B

The refrigerant used in steam jet cooling is:

- A. Steam
- B. R-11
- C. Ammonia
- D. Water

Ans: D

The total heat of the air is a function of

- A. WB temperature
- B. DP temperature
- C. DB temperature
- D. WB depression

Ans: A

Boiling point of Freon-12 at atmospheric pressure is:

- A. 21 deg F
- B. 15 deg F
- C. 5 deg F
- D. 28 deg F

Ans: A

Which of the following is NOT a type of water cooled condenser in refrigeration?

- A. Double pipe
- B. Double shell
- C. Shell and coil
- D. Shell and tube

Ans: B

Component of absorption refrigeration system in which the solution is cooled by cooling water.

- A. Rectifier
- B. Generator
- C. Evaporator
- D. Absorber

Ans: D

Cascade refrigeration cycle is often used in industrial process where objects must be cooled to temperature below:

- A. -46 deg C
- B. -56 deg C
- C. -66 deg C
- D. -76 deg C

Ans: A

Type of refrigerant control designed to maintain a pressure difference while the compressor is operating.

- A. Thermostatic expansion valve
- B. Using low side float flooded system
- C. Automatic expansion valve
- D. Capillary tube

Ans: D

As a rule of thumb, for a specified amount of compressed air, the power consumption of the compressor decreases by _____ for each 3 deg C drop in the temperature inlet air to the compressor.

- A. 1 percent
- B. 1.5 percent
- C. 2 percent
- D. 2.5 percent

Ans: A

Modern way of detecting air compressor leak is by using

- A. Soap and water
- B. Air leak detector
- C. Acoustic leak detector
- D. Ammonia leak detector

Ans: C

For foundation of stacks, the maximum pressure on the soil is equal to the pressure due to the weight and the _____.

- A. Soil movement
- B. Wind movement
- C. Ground movement
- D. Engine movement

Ans: B

Foundation bolts of specified size should be used and surrounded by a pipe sleeve with an inside diameter of at least

- A. 3 times the diameter of engine bolt
- B. 2 times the diameter of engine bolt
- C. 3 times the diameter of anchor bolt
- D. 2 times the diameter of anchor bolt

Ans: B

For multi stage compression of an ideal Brayton cycle, the back ratio will

- A. Increase
- B. Decrease
- C. Remains the same
- D. None of these

Ans: B

Type of turbine that has a specific speed below 5.

- A. Impulse turbine
- B. Propeller turbine
- C. Francis turbine
- D. Deriaz turbine

Ans: A

A high discharge type of turbine

- A. Impulse turbine
- B. Propeller turbine
- C. Francis turbine
- D. Deriaz turbine

Ans: C

Use to minimize the speed rise due to a sudden load rejection

- A. Needle valve
- B. Wicket gate shut-off
- C. Shut-off valve
- D. Jet deflector

Ans: D

Is the speed of a turbine when the head on the turbine is one meter

- A. Specific speed
- B. Rated speed
- C. Utilized speed
- D. Unit speed

Ans: D

Is a fluid property which refers to the intermolecular attraction by which the separate particles of the fluid are held together.

- A. Cohesion
- B. Adhesion
- C. Surface tension
- D. Hypertension

Ans: A

Which of the following is NOT the cause of black smoke in diesel engine?

- A. Fuel valve open too long
- B. High compression pressure
- C. Carbon in exhaust pipe
- D. Overload on engine

Ans: B

Which of the following is not a method of starting a diesel engine?

- A. Manual: rope, crank and kick
- B. Electric (battery)
- C. Compressed air
- D. Using another generator

Ans: D

Two-stroke engine performs _____ to complete one cycle.

- A. Suction and discharge stroke
- B. Compression and power stroke
- C. Power and exhaust stroke
- D. Suction and exhaust stroke

Ans: B

A type of geothermal plant used when there is a presence of brine extracted from underground

- A. Dry geothermal plant
- B. Double-flash geothermal
- C. Single flash geothermal plant
- D. Binary geothermal plant

Ans: D

Is the most important safety device on the power boiler.

- A. Check valve
- B. Gate valve
- C. Safety valve
- D. Globe valve

Ans: C

During hydrostatic test, the safety valves should be

- A. Removed
- B. Open
- C. Closed
- D. Partially closed

Ans: A

Where deaerating heaters are not employed, it is recommended that the temperature of the feed water be not less than _____

- A. 197 deg C
- B. 102 deg C
- C. 104 deg C
- D. 106 deg C

Ans: A

Is a reaction during which chemical energy is released in the form of heat.

- A. Cosmic reaction
- B. Ethnic reaction
- C. Endothermic reaction
- D. Exothermic reaction

Ans: D

By reheating the steam in an ideal Rankine cycle the heat rejected will.

- A. Increase
- B. Decrease
- C. Remains the same
- D. None of these

Ans: A

By increasing the boiler pressure in Rankine cycle the moisture content at boiler exit will.

- A. Increase
- B. Decrease
- C. Remains the same
- D. None of those

Ans: A

Presently the highest steam temperature allowed at the turbine inlet is about _____.

- A. 340 deg C
- B. 520 deg C
- C. 620 deg C
- D. 1020 deg C

Ans: C

Two most common gases employed in Stirling and Ericsson cycles are.

- A. Air and helium
- B. Oxygen and helium
- C. Hydrogen and helium
- D. Nitrogen and helium

Ans: C

In most common design of gas turbine, the pressure ratio ranges from

- A. 10 to 12
- B. 11 to 16
- C. 12 to 18
- D. 15 to 20

Ans: B

In brayton cycle, the heat is transformed during what process?

- A. Constant temperature
- B. Isentropic process
- C. Isobaric process
- D. Isochoric process

Ans. C

The fuel injection process in diesel engine starts when the piston _____

- A. Is at the TDC
- B. Leaving TDC
- C. Approaches TDC
- D. Halfway of the stroke

Ans. C

If the cut-off ratio of diesel cycle increases, the cycle efficiency will

- A. Decrease
- B. Increase
- C. Remains the same
- D. None of these

Ans. A

The fuel used in a power plant that is used during peak periods.

- A. Gas
- B. Solid
- C. Liquid
- D. None of these

Ans. C

Typical compression ratio of Otto cycle is

- A. 6
- B. 8
- C. 10
- D. 12

Ans. B

If joule Thomson coefficient is equal to zero, then the process will become

- A. Isentropic
- B. Isenthalpic
- C. Isobaric
- D. Isothermal

Ans. D

If the fluid passed through a nozzle its entropy will:

- A. Increase
- B. Decrease
- C. Remains the same
- D. None of these

Ans. C

Refrigerant consisting of mixtures of two or more different chemical compounds, often used individuals as refrigerant for other applications.

- A. Suspension
- B. Compound reaction
- C. Blends
- D. Mixing of refrigerant

Ans. C

Pairs of mating stop valves that allow sections of a system to be joined before opening these valves or separated after closing them

- A. Check valve
- B. Gate valve
- C. Safety valve
- D. Companion valve

Ans. D

An enclosed passage way that limits travel to a single path.

- A. Corridor
- B. Hallway
- C. Lobby
- D. Tunnel

Ans. A

For immediate dangerous to life or health (IDLH), the maximum concentration from which unprotected persons are able to escape within _____ without escape-impairing symptoms or irreversible health.

- a. 15 min
- b. 1 min
- c. 20 min
- d. 30 min

Ans. D

The volume as determined from internal dimensions of the container with no allowance for the volume of internal parts.

- a. Internal allowance
- b. Internal gross volume
- c. Internal interference volume
- d. Internal fits volume

Ans B

A waiting room or large hallway serving as a waiting room

- a. Terrace
- b. Rest room
- c. Compound room
- d. Lobby

Ans. d

A continuous and unobstructed path of travel from any point in a building or structure to a public way

- a. Average of aggress
- b. Mean of aggress
- c. Hallway of aggress
- d. Pathway of aggress

Ans. B

Any device or portion of the equipment used to increase refrigerant pressure

- a. Pressure relief device
- b. Pressure-imposing element
- c. Pressure lift device
- d. Pressure limiting device

Ans B

The quantity of refrigerant stored at some point is the refrigeration system for operational, service, or standby purposes/

- a. Pressure vessel
- b. Pumpdown charge
- c. Liquid receiver
- d. Accumulator

Ans B

Secondary refrigerant is a liquid used for the transmission of heat, without a change of state, and having no flash point or a flash point above _____ as determined from ASTM

- a. 150 deg F
- b. 160 F
- c. 180 F
- d. 200 F

Ans A

A service valve for dual pressure-relief devices that allows using one device while isolating the other from the system, maintaining one valve in operation at all times.

- a. Three-way valve
- b. Two-way valve
- c. One-way valve
- d. Four-way valve

Ans A

Tubing that is enclosed and therefore exposed to crushing, abrasion, puncture, or similar damage after installation.

- a. Protected tubing
- b. Bare tubing
- c. Open tubing
- d. Unprotected tubing

Ans D

Refers to blends comprising multiple components of different volatile that, when used in refrigeration cycles, change volumetric composition and saturation temperature as they evaporate (boil) or condense at constant pressure.

- a. Zeolite
- b. Blending
- c. Composition
- d. Zeotropic

Ans D

Is a premises or that of a premise from which, because they are disabled, debilitated, or confined, occupants cannot readily leave without the assistance of others

- a. Institutional occupancy
- b. Public assembly occupance
- c. Residential occupancy
- d. Commercial occupancy

Ans A

Is one in which a secondary coolant is in direct contact with the air or other substance to be cooled or heated.

- a. Double indirect open spray system
- b. Indirect open spray system
- c. Indirect closed system
- d. Indirected, verted closed system

Ans B

Refrigerant number R-744 is

- a. Butane
- b. Carbon monoxide
- c. Propane
- d. Carbon dioxide

Ans D

Refrigerant number R-1150 is

- a. Propylene
- b. Ethane
- c. Ethane
- d. Methyl formate

Ans B

Refrigerant R-40 is

- a. Chlorodifluoromethane
- b. Difluoromethane
- c. Ammonia
- d. Chloromethane

Ans D

When the air duct system serves several enclosed spaces, the permissible quantity of refrigerant in the system shall not exceed the amount determined by using the total volume of those spaces in the which the

airflow cannot be reduced to less than ____ of its maximum when the fan is operating

- a. One-quarter
- b. One half-quarter
- c. Three-quarter
- d. One-fourth-quarter

Ans A

The space above a suspended ceiling shall not be included in calculating the permissible quantity of refrigerant in the system unless such space is continuous and is part of the air return system

- a. Partition
- b. Plenums
- c. Separator
- d. Plate divider

Ans B

Which of the following is not a possible location of service valve?

- a. Suction of compressor
- b. Discharge of compressor
- c. Outlet of liquid receiver
- d. Outlet of condenser

Ans D

A coil in series with evaporator that is use to prevent the liquid refrigerant entering the compressor

- a. Accumulator
- b. Liquid superheater
- c. Drier loop
- d. Liquid suction heat exchanger

Ans C

A type of valve connected from discharge of compressor directly to suction that is normally closed and will open automatically only if there is high discharge pressure

- a. Check valve
- b. Solenoid valve
- c. King valve
- d. Relief valve

Ans B

Use to increase the capacity of condenser

- a. Water regulating valve
- b. Desuperheating coils
- c. Liquid-suction heat exchanger
- d. Condenser heating coils

Ans B

Is use to subcooled the refrigerant from the condenser

- a. Liquid line
- b. Condenser subcooler
- c. Desuperheating coils
- d. Liquid receiver

ans A

Which of the following is NOT a part of low pressure side in refrigeration system?

- a. Compressor
- b. Condenser
- c. Liquid line
- d. Suction line

Ans D

Which of the following is NOT a part of condensing unit?

- a. Compressor

- b. Discharge line
- c. Condenser
- d. Liquid line

Ans.D

By subcooling the refrigerant in refrigeration system, the compressor power per unit mass will

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans C

Superheating the refrigerant in refrigeration system, the specific volume at compressor suction will

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans C

By subcooling the refrigerant in refrigeration system, the specific volume at compressor suction will

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans c

Pressure loss due to friction at the condenser, the compressor power per unit mass will

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans c

Which of the following is NOT a type of air-cooled condenser?

- a. Shell and tube
- b. Natural draft
- c. Forced draft
- d. Induced draft

Ans a

A type of refrigerant control typically used in household refrigeration

- a. Thermostatic expansion valve
- b. Automatic expansion valve
- c. Capillary tube
- d. High side float

Ans C

Type of condenser that operates like a cooling water

- a. Air-cooled condenser
- b. Evaporative condenser
- c. Shell and tube condenser
- d. Water cooled condenser

Ans b

The major problem of heat pump is

- a. Refrigerant used
- b. Outside air
- c. Supply air
- d. Frosting

Ans D

Dominant refrigerant used in commercial refrigeration system

- a. R11
- b. R22
- c. R12
- d. R502

Ans D

Cascade refrigerant system are connected in

- a. Series
- b. Parallel
- c. Series-parallel
- d. Parallel

Ans A

Is use to heat up the solution partially before entering the generator in absorption refrigeration system

- a. Rectifier
- b. Absorber
- c. Regenerator
- d. Pump

ans C

The COP of actual absorption refrigeration system is usually

- a. Less than 1
- b. Less than 2
- c. Less than 3
- d. Less than 4

Ans A

Sight glass is often located at

- a. Discharge line
- b. Liquid line
- c. Between condenser and liquid receiver
- d. Suction line

Ans B

Use to detects a vibration in current caused by the ionization of decomposed refrigerant between two opposite charged platinum electrodes

- a. Electronic detector
- b. Halide torch
- c. Bubble method
- d. Pressurizing

Ans a

The ability of oil to mix with refrigerants

- a. Carbonization
- b. Purging
- c. Mixing
- d. Miscibility

Ans D

Joints and all refrigerants-containing parts of a refrigerating system located in an air duct carrying conditioned air to and from an occupied space shall be constructed to withstand a temperature of ____ without leakage into the airstream.

- a. 550F
- b. 600F
- c. 650F
- d. 700F

Ans D

Refrigerant piping crossing an open space that affords passageway in any building shall be not less than _____ above the floor unless the piping is

located against the ceiling of such space and is permitted by the authority having jurisdiction

- a. 2.2 m
- b. 3.2 m
- c. 4.2 m
- d. 5.2 m

Ans A

Methyl chloride shall not be in contact with

- a. Aluminum
- b. Zinc
- c. Magnesium
- d. All of these

Ans D

Shall not be in contact with any halogenated refrigerants

- a. Aluminum
- b. Zinc
- c. Magnesium
- d. All of these

Ans C

Are suitable for use in ammonia system

- a. Copper
- b. Aluminum and its alloy
- c. Plastic
- d. Cast iron

Ans B

In a pressure-relief device is used to protect a pressure vessel having an inside dimension of 6 in or less, the ultimate strength of the pressure vessel so protected shall be sufficient to withstand a pressure at least ____ the design pressure

- a. 2 times
- b. 3 times
- c. 4 times
- d. 5 times

Ans B

Seats and discs shall be limited in distortion, by pressure or other cause, to set pressure change of not more than ____ in a span of five years.

- a. 1%
- b. 5%
- c. 10%
- d. 50%

Ans B

Liquid receivers, if used or parts of a system designed to receive the refrigerant charge during pumpdown charge. The liquid shall not occupy more than ____ of the volume when temperature of the refrigerant is 90F

- a. 80%
- b. 85%
- c. 90%
- d. 95%

Ans C

The discharge line (B4) shall be vented to the atmosphere through a ____ fitted to its upper extremity.

- a. Nozzle
- b. Convergent-divergent nozzle
- c. Pipe
- d. Diffuser

Ans D

Convert fossil fuels into the shaft work

- a. Nuclear power plant
- b. Gas turbine power plant
- c. Dendrothermal power plant
- d. Thermal power plant

Ans d

Ultimate strength drops by 30% as steam temperature raises from ____ for unalloyed steel

- a. 300 to 400 c
- b. 400 to 500 c
- c. 600 to 700 c
- d. 700 to 800 c

Ans B

Recent practice limits steam temperature to

- a. 438 c
- b. 538 c
- c. 648 c
- d. 738 c

Ans a

In a closed feed water heater, the feed water pass through

- a. Inside the tube
- b. Outside the tube
- c. Inside the shell
- d. Outside the shell

Ans a

Is use if extracted steam upon condensation gets subcooled

- a. Trap
- b. Deaerator
- c. Filter
- d. Drain cooler

Ans D

Needs only single pump regardless of number of heaters

- a. Open heater
- b. Closed heater
- c. Mono heater
- d. Regenerative heater

Ans B

Is also known as deaerator

- a. Open heater
- b. Closed heater
- c. Reheat heater
- d. Regenerative heater

Ans a

Dissolve gases like ____ makes water corrosive react with metal to form iron oxide

- a. O2 and N2
- b. O2 and CO
- c. O2 and CO2
- d. N2 and SO2

Ans C

A cycle typically used in paper mills, textile mills, chemical factories, sugar factories and rice mills

- a. Cogeneration cycle
- b. Combined cycles
- c. By-products cycle
- d. Cascading cycle

Ans a

When process steam is basic need and power is byproduct, this cycle is known as

- a. Cogeneration cycle
- b. Combined cycle
- c. By-product cycle
- d. Cascading cycle

Ans C

A type of turbine employed where steam continuously extracted for process heating

- a. Back-pressure turbine
- b. Gas turbine
- c. Steam turbine
- d. Passout turbine

Ans D

Which of the following is used for binary cycle power generation for high temperature application?

- a. Mercury
- b. Sodium
- c. Potassium
- d. All of these

Ans D

Critical temperature of mercury is

- a. 1160 c
- b. 1260 c
- c. 1360 c
- d. 1460 c

Ans D

Critical pressure of mercury is

- a. 100 MPa
- b. 108 MPa
- c. 128 MPa
- d. 158 MPa

Ans B

Method used in converting heat directly to electricity by magnetism

- a. Electromagnetic induction
- b. Magnetodynamic
- c. Magnetohydrodynamic
- d. Thermoelectric

Ans C

Which of the following is not a material used for thermoelectric elements

- a. Bismuth telluride
- b. Lead telluride
- c. Zinc telluride
- d. Germanium

ans C

a type of coal formed after anthracite

- a. Lignite
- b. Bituminous
- c. Peat
- d. Graphite

Ans D

Which of the following is lowest grade of coal?

- a. Peat
- b. Lignite
- c. Sub-bituminous
- d. Bituminous

ansB

which of the following helps in the ignition of coal?

- a. Moisture
- b. Ash
- c. Fixed carbon
- d. Volatile matter

Ans D

Is the ratio of fixed carbon and volatile matter.

- A. Air-fuel ratio
- B. Fuel ratio
- C. Combustion ratio
- D. Carbon-volatile ratio

Ans. B

A suspension of a finely divide fluid in another.

- A. Filtration
- B. Floatation
- C. Emulsion
- D. Separation

Ans. C

Contains 90% gasoline and 10% ethanol.

- A. Gasohol
- B. Gasonol
- C. Gasothanol
- D. Gasethanol

Ans. A

Process used commercially in coal liquefaction.

- A. Tropsch process
- B. Fisher process
- C. Fisher-tropsch process
- D. Mitch-tropsch process

Ans. C

Is an organic matter produced by plants in both land and water.

- A. Bio-ethanol
- B. Biomass
- C. Petroleum
- D. Biogradable

Ans. B

In thermal power plant, induced draft fans are located at the

- A. Exit of furnace
- B. Foot of the stack
- C. Above the stock
- D. Top of the stack

Ans. B

In thermal power plant, forced draft fans are installed at the

- A. Foot of the stack
- B. Top of the stack
- C. Exit of the preheater
- D. Inlet of the preheater

Ans. D

Known as drum less boiler.

- A. La Mont boiler
- B. Fire tube boiler
- C. Force circulation boiler
- D. Once-through boiler

Ans. D

Reduces the steam temperature by spraying low temperature water from boiler drum.

- A. Reheater
- B. Preheater
- C. Desuperheater
- D. Superheater

Ans. C

Carbon dioxide can be removed by:

- A. Deaeration
- B. Aeration
- C. Evaporation
- D. Vaporization

Ans. B

Is often used to absorb silica from water.

- A. Sorbent
- B. Rectifier
- C. Silica gel
- D. Magnesium hydroxide

Ans. D

Presence of excess hydrogen ions makes the water

- A. Acidic
- B. Alkalinity
- C. Base
- D. Hydroxycity

Ans. A

PH of water varies with

- A. Pressure
- B. Temperature
- C. Density
- D. Volume

Ans. B

Ph value of _____ is usually maintained for boiler water to minimized corrosion.

- A. 8.5
- B. 9.5
- C. 10.5
- D. 11.5

Ans. C

What type of turbine that has a degree of reaction of $\frac{1}{2}$?

- A. Impulse turbine
- B. Reaction turbine
- C. Rarsons turbine
- D. Deriaz turbine

Ans. C

Tranquil flow must always occur

- A. Above the normal depth
- B. Above the critical depth
- C. Below the normal depth
- D. Below the critical depth

Ans. B

Which of the following head loss coefficient among the following types of entrance?

- A. Bell mouth
- B. Square edge
- C. Reentrant
- D. It depends

Ans. C

What waste treatment method involves of algae from stabilization pond effluents?

- A. Sedimentation
- B. Floatation
- C. Filtration
- D. Microscreening

Ans. D

The number of nozzles will depend on the quantity of steam required by the turbine. If nozzles occupy the entire the ring, the turbine is said to have:

- A. Partially full peripheral admission
- B. One-half peripheral admission
- C. Maximum peripheral admission
- D. Full peripheral admission

Ans. D

Tandem compound units may also have two low-pressure castings that produces:

- A. Single flow
- B. Double flow
- C. Triple flow
- D. Quadruple flow

Ans. C

A type of turbine used for driving pumps, fans, and other auxiliaries in power plant commonly operate at exhaust pressure approximating atmospheric.

- A. Tandem compound turbine
- B. Passout turbine
- C. Cross-compound turbine
- D. Back pressure turbine

Ans. D

A governor with 0% regulation is termed as:

- A. Isochronous governor
- B. Synchronous governor
- C. Isenchronous governor
- D. Isobarnous governor

Ans. A

The speed regulation for most turbine-generators is adjustable from:

- A. 2 to 6%
- B. 4 to 8%
- C. 6 to 10%
- D. 8 to 12%

Ans. A

Poppet valves of steam turbine are used for extraction pressure of:

- A. 20 to 120 psig
- B. 20 to 150 psig
- C. 20 to 130 psig
- D. 20 to 140 psig

Ans. B

When both bearings of steam engines are on one side of the of the connecting rod, the engine is referred to as:

- A. Center-crank engine
- B. Side crank engine
- C. Under crank engine
- D. Standard crank engine

Ans. B

When the valve in steam engine is in mid-position of its travel, it will cover the steam port by an amount known

- A. Steam lap
- B. Partial lap
- C. Full lap
- D. Angular lap

Ans. A

A type of governor in steam engine that do not control the actual admission of steam to the cylinder but controls the pressure of the steam.

- A. Flyball governor
- B. Variable cut-off governor
- C. Throttling governor
- D. Shaft governor

Ans. C

By inter-cooling using two stage compressor of Brayton cycle, the backwork ratio will:

- A. Increase
- B. Decrease
- C. Remains constant
- D. None of these

Ans. B

On dynamic similitude, the relation which represents the ratio of inertia force to pressure force is:

- A. Froude number
- B. Cauchy number
- C. Euler number
- D. Reynolds number

Ans. C

What is the maximum velocity in a sewer flowing full?

- A. 0.6 m/sec
- B. 0.9 m/sec
- C. 1.2 m/sec
- D. 1.8 m/sec

Ans. A

A temporary structure constructed to exclude water from the side of the foundation during its excavation and construction is called.

- A. Caisson
- B. Retaining wall
- C. Cofferdam
- D. Earth dam

Ans. C

Which is not a physical characteristic of water?

- A. Total suspended and dissolved solids
- B. Turbidity
- C. Color
- D. Hardness

Ans. D

Which dam is best for weak foundation?

- A. Gravity
- B. Arch
- C. Buttress
- D. Earth

Ans. C

What is the volume of water which will drain freely from the aquifer?

- A. Specific yield
- B. Reservoir yield
- C. Safe yield
- D. Secondary yield

Ans. A

What is the line defined by the water level in a group of artesian wells?

- A. Water table
- B. Piezometric surface
- C. Specific yield
- D. All of the above

Ans. B

Select the one that is a positive indication of pollution of a river.

- A. Acidity
- B. Oxygen content
- C. Chloride content
- D. Nitrite content

Ans. C

The cooling water is made to fall in series of baffles to expose large surface area for steam led from below to come in direct contact.

- A. Spray condenser
- B. Surface condenser
- C. Jet condenser
- D. Barometric condenser

Answer: D

Show the variation of river flow (discharge) with time

- A. Hydrograph
- B. Hyetograph
- C. Mass curve
- D. Flow duration curve

Answer: A

Is an open channel erected on a surface above the ground.

- A. Canal
- B. Tunnel
- C. Pentstock
- D. Flume

Answer: D

Type of turbine used up to 300 m head

- A. Impulse turbine
- B. Francis turbine
- C. Propeller turbine
- D. Deriaz turbine

Answer: D

Oil is atomized either by air blast or pressure jet at about

- A. 60 bar
- B. 70 bar
- C. 80 bar
- D. 90 bar

Answer: B

Type of solid injection that use single pump supplies fuel under high pressure to a fuel header.

- A. Common rail injection
- B. Individual pump injection system
- C. Distributor system

D. Single rail injection

Answer: A

Water flow in diesel engine that is caused by density differential.

- A. Thermosiphon cooling
- B. Thermostat cooling
- C. Pressure water cooling
- D. Evaporative cooling

Answer: A

Typo of lubrication system in diesel engine in which oil from pump is carried to a separate storage tank outside the engine cylinder and used for high capacity engine.

- A. Mist lubrication system
- B. Wet sump lubrication
- C. Splash system
- D. Dry sump lubrication system

Answer: D

Produces extreme pressure differentials and violent gas vibration.

- A. Vibration
- B. Detonation
- C. Explosion
- D. Knocking

Answer: B

In a spark ignition engine, detonation occurs near the _____.

- A. End of combustion
- B. Middle combustion
- C. Beginning of combustion
- D. Beginning of interaction

Answer: A

In a compression ignition engine, the detonation occurs near the _____.

- A. End of combustion
- B. Middle of combustion
- C. Beginning of combustion
- D. Beginning of interaction

Answer: C

Morse test is used to measure the _____ of multi-cylinder engine.

- A. Break power
- B. Indicated power
- C. Friction power
- D. Motor power

Answer: B

Ignition delay can be minimized by adding _____ to decrease engine knocking.

- A. Ethyl ether
- B. Ethyl chloride
- C. Ethyl nitrate
- D. Ethyl oxide

Answer: C

At any point in fluid at rest, the pressure is the same in all directions. This principle is known as:

- A. Bernoulli principle
- B. Archimedes principle
- C. Pascal's law
- D. Torricelli's law

Answer: C

The hot-wire manometer is used to measure

- A. Pressure in gases
- B. Pressure in fluids

C. Wind velocities at airports

D. Gas velocities

Answer: D

The pitot static tube measures

- A. The static pressure
- B. The gage pressure
- C. The total pressure
- D. The dynamic pressure

Answer: A

The terminal velocity of a small sphere settling in a viscous fluid varies as the

- A. First power of its diameter
- B. Inverse of fluid viscosity
- C. Inverse square of the diameter
- D. Inverse of the diameter

Answer: B

Pressure drag results from

- A. Skin friction
- B. Deformation drag
- C. Breakdown of potential flow near the forward stagnation point
- D. Occurrences of wake

Answer: A

The pressure coefficient is the ratio of pressure forces to:

- A. Viscous forces
- B. Inertia forces
- C. Gravity forces
- D. Surface tension force

Answer: B

Which instrument is used to measure humidity of the atmosphere continuously?

- A. Barograph
- B. Thermograph
- C. Hydrograph
- D. Thermo-hydrograph

Answer: C

Entrance losses between tank and pipe or losses through elbows, fittings and valves are generally expressed as a function of

- A. Kinetic energy
- B. Pipe diameter
- C. Friction factor
- D. Volume flow rate

The air that contains no water vapor is called

- A. Zero air
- B. Saturated air
- C. Dry air
- D. Humid air

Answer: C

In psychrometric chart, the constant-enthalpy lines coincide with constant-temperature lines at temperature

- A. Above 50°C
- B. Below 40°C
- C. Below 50°C
- D. Above 10°C

Answer: C

The amount of moisture in air depends on its

- A. Pressure
- B. Volume

- C. Temperature
- D. Humidity

Answer: C

The deep body temperature of healthy person is maintained constant at

- A. 27°C
- B. 37°C
- C. 47°C
- D. 48°C

Answer: B

Air motion also plays important role in

- A. Surroundings
- B. Cooling
- C. Human comfort
- D. None of these

Answer: C

During simple heating and cooling process has a ____ humidity ratio

- A. Increasing
- B. Decreasing
- C. Constant
- D. None of these

Answer: C

The ____ follows a line of constant wet-bulb temperature on the psychrometric chart.

- A. Evaporative cooling process
- B. Condensive cooling process
- C. Direct cooling process
- D. None of these

Answer: A

A vapor which is not about to condensate is called a

- A. mixture of vapor and liquid
- B. critical vapor
- C. Superheated vapor
- D. None of these

Answer: C

Passing from the solid phase directly into vapor phase is called

- A. Condensation
- B. Fusion
- C. Sublimation
- D. None of these

Answer: C

Robert Boyle observed during his experiments with a vacuum chamber that the pressure of gases is inversely proportional to their

- A. Temperature
- B. Pressure
- C. Volume
- D. None of these

Answer: C

____ is energy in transition

- A. Heat
- B. Work
- C. Power
- D. None of these

Answer: A

Is the mode of energy transfer between a solid surface and the adjacent liquid or gas which is in motion, and it involves combine effects of conduction and fluid motion.

- A. Conduction
- B. Convection
- C. Radiation
- D. None of these

Answer: B

Radiation is usually considered as

- A. Surface phenomenon
- B. Surface interaction
- C. Surface corrosion
- D. None of these

Answer: A

Work is ____ between the system and the surroundings.

- A. Work interaction
- B. Energy interaction
- C. Heat interaction
- D. None of these

Answer: B

Is a process during which the system remains in equilibrium at all times

- A. Quasi-equilibrium
- B. Static equilibrium
- C. Dynamic equilibrium
- D. None of these

Answer: A

In the absence of any work interactions between a system and its surroundings , the amount of net heat transfer is equal

- A. The change in total energy of a closed system
- B. To heat and work
- C. Energy interactions
- D. none of these

Answer: A

The constant volume and constant pressure specific heats are identical for

- A. compressible substance
- B. incompressible substance
- C. compressible gas
- D. none of these

Answer: B

The velocity of fluid is zero at wall and maximum at the center because of the

- A. velocity effect
- B. viscous effect
- C. temperature effect
- D. none of these

Answer: B

For steady flow devices, the volume of the control volume is

- A. increase
- B. decrease
- C. constant
- D. none of these

Answer: C

Work done in turbine is ____ since it is done by the fluid.

- A. Positive
- B. Negative
- C. Zero
- D. None of these

Answer: A

Reheating process in Brayton cycle, the turbine work will

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans. A

Which of the following is the chemical formula of ethanol?

- a. C_7H_{16}
- b. C_2H_6O
- c. C_7H_8
- d. C_6H_{12}

Ans. B

Which of the following is the chemical formula of heptane?

- a. C_7H_{16}
- b. C_2H_6O
- c. C_7H_8
- d. C_6H_{12}

Ans. A

Which of the following is the chemical formula of hexane?

- a. C_7H_{16}
- b. C_2H_6O
- c. C_7H_8
- d. C_6H_{12}

Ans. D

Which of the following is the chemical formula of toluene?

- a. C_7H_{16}
- b. C_2H_6O
- c. C_7H_8
- d. C_6H_{12}

Ans. C

As the air passes through a nozzle, which of the following will increase?

- a. Temperature
- b. Enthalpy
- c. Internal energy
- d. Mach number

Ans. D

As the air passes through a diffuser, which of the following will decrease?

- a. Temperature
- b. Enthalpy
- c. Internal energy
- d. Mach number

Ans. D

As the air passes through a nozzle, which of the following will decrease?

- a. Temperature
- b. Enthalpy
- c. Internal energy
- d. Mach number

Ans. C

As the air passes through a diffuser, which of the following will increase?

- a. Density
- b. Entropy
- c. Mach number
- d. Velocity

Ans. A

As the air passes through a diffuser, which of the following will NOT be affected?

- a. Density
- b. Entropy
- c. Mach number
- d. Velocity

Ans. B

After passing through a convergent-divergent nozzle the temperature of air will:

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans. B

After passing through a convergent-divergent nozzle the density of air will:

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans. B

After passing through a convergent-divergent nozzle the mach number of air will:

- a. Increase
- b. Decrease
- c. Remains the same
- d. None of these

Ans. B

By increasing the temperature source of Carnot cycle, which of the following will not be affected?

- a. Efficiency
- b. Work
- c. Heat added
- d. Heat rejected

Ans. D

By decreasing the temperature source of Carnot cycle, which of the following will not be affected?

- a. Efficiency
- b. Work
- c. Heat added
- d. Heat rejected

Ans. C

By superheating the refrigerant in vapor compression cycle with useful cooling, which of the following will increase? (use per unit mass analysis)

- a. Condenser pressure
- b. Evaporator pressure
- c. Quality after expansion
- d. Heat rejected from condenser

Ans. D

By superheating the refrigerant in vapor compression cycle with useful cooling, which of the following will decrease? (use per unit mass analysis)

- a. Refrigerated effect
- b. COP
- c. Compressor power
- d. Mass flow rate

Ans. D

By superheating the refrigerant in vapor compression cycle without useful cooling, which of the following will decrease? (use per unit mass analysis)

- a. Heat rejected
- b. COP
- c. Compressor power
- d. Specific volume at suction

Ans. B

By superheating the refrigerant in vapor compression cycle without useful cooling, which of the following will increase? (use per unit mass analysis)

- a. Heat rejected
- b. COP
- c. Compressor power
- d. Specific volume at suction

Ans. C

By superheating the refrigerant in vapor compression cycle without useful cooling, which of the following will not be affected? (use per unit mass analysis)

- a. Refrigerated effect
- b. COP
- c. Compressor power
- d. Mass flow rate

Ans. C

By sub-cooling the refrigerant in vapor compression cycle at condenser exit, which of the following will increase? (use per unit mass analysis)

- a. Refrigerated effect
- b. Specific volume at suction
- c. Compressor power
- d. Mass flow rate

Ans. A

By sub-cooling the refrigerant in vapor compression cycle at condenser exit, which of the following will decrease? (use per unit mass analysis)

- a. Refrigerated effect
- b. Specific volume at suction
- c. Compressor power
- d. Mass flow rate

Ans. D

By increasing the vaporizing temperature in vapor compression cycle, which of the following will increase? (Use per unit mass analysis)

- A. mass flow rate
- B. COP
- C. specific volume at suction
- D. compressor work

Answer: B

By increasing the vaporizing temperature in vapor compression cycle, which of the following will decrease? (Use per unit mass analysis)

- A. Refrigeration effect
- B. COP
- C. evaporator temperature
- D. Temperature difference between evaporator and compressor

Answer: D

By increasing the condenser pressure in vapor compression cycle, which of the following will increase? (Use per unit mass analysis)

- A. moisture content after expansion
- B. compressor power
- C. heat rejected from condenser
- D. mass flow rate

Answer: A

If the pressure drop in the condenser Increases in a vapor compression cycle, which of the following will increase? (Use per unit mass analysis)

- A. mass flow rate
- B. compressor power
- C. heat rejected in the condenser
- D. specific volume of suction

Answer: C

If the pressure drop in the condenser increases in a vapor compression cycle, which of the following will decrease? (Use per unit mass analysis)

- A. Refrigeration effect
- B. mass flow rate
- C. heat rejected in the condenser
- D. compressor power

Answer: B

If the pressure drop in the condenser increases in a vapor compression cycle, which of the following will not be affected? (Use per unit mass analysis)

- A. compressor power
- B. mass flow rate
- C. heat rejection in the condenser
- D. COP

Answer: A

If the pressure drop in the evaporator increases in a vapor compression cycle, which of the following will increase? (Use per unit mass analysis)

- A. Refrigerating effect
- B. vaporizing temperature
- C. heat rejected in the condenser
- D. COP

Answer: C

If the pressure drop in the evaporator increases in a vapor compression cycle, which of the following will decrease? (Use per unit mass analysis)

- A. specific volume at suction
- B. compressor power
- C. heat rejected in the condenser
- D. COP

Answer: D

By lowering the condenser pressure in Rankine cycle, which of the following will decrease? (Use per unit mass analysis)

- A. pump work
- B. turbine
- C. heat rejected
- D. cycle efficiency

Answer: C

By increasing the boiler pressure in Rankine cycle, which of the following will decrease? (Use per unit mass analysis)

- A. heat rejected
- B. pump work
- C. cycle efficiency
- D. moisture

Answer: A

By superheating the steam to a higher temperature in Rankine cycle, which of the following will decrease? (Use per unit mass analysis)

- A. moisture content at the turbine exhaust
- B. turbine work
- C. heat added
- D. heat rejected

Answer: A

By superheating the steam to a higher temperature in Rankine cycle, which of the following will increase? (Use per unit mass analysis)

- A. moisture content at the turbine exhaust
- B. pump work
- C. condenser pressure
- D. cycle efficiency

Answer:

By reheating the steam before entering the second stage in Rankine cycle, which of the following will decrease?

- A. turbine work
- B. moisture content after expansion
- C. heat added
- D. heat rejected

Answer: B

When Rankine cycle is modified with regeneration, which of the following will increase?

- A. turbine work
- B. heat added
- C. heat rejected
- D. cycle efficiency

Answer: D

Is the combination of base load and peaking load.

- A. rated load
- B. intermediate load
- c. combine load
- D. over-all load

Answer: B

Sum of the maximum demand over the simultaneous maximum demand.

- A. use factor
- B. capacity factor
- C. demand factor
- D. diversity factor

Answer: D

Regenerative with feed heating cycle with infinite number of feedwater heaters thus efficiency is equal to:

- A. otto cycle
- B. stirling cycle
- C. erricson cycle
- D. carnot cycle

Answer: D

A type of turbine used in desalination of sea water.

- A. back pressure turbine
- B. passout turbine
- C. peaking turbine
- D. reaction turbine

Answer: A

States that when conductor and magnetic field move relatively to each other, an electric voltage is induced in the conductor.

- A. Maxwell's law
- B. Kirchoff's law
- C. Faraday's law
- D. Newtons law

Answer:C

Transfers heat directly to electrical energy by utilizing thermionic emissions.

- A. Thermionic motor
- B. Thermionic generator
- C. Thermionic converter

D. Thermionic cell

Answer:B

Is the largest group of coal containing 46-86% of fixed carbon and 20 to 40% volatile matter.

- A. anthracite
- B. sub-anthracite
- C. Bituminous
- D. Sub-bituminous

Ans. C

When 1 gram of coal is subjected to a temperature of about 105°C for a period of 1 hour, the loss in weight of the sample gives the:

- A. volatile matter
- B. ash
- C. Fixed carbon
- D. moisture content

Answer:D

When 1 gram of sample of coal is placed in a crucible and heated 950°C and maintain at that temperature for 7 minutes there is a loss in weight due to elimination of:

- A. volatile matter and moisture
- B. ash
- C. Fixed carbon
- D. moisture content

Answer:A

Consist of hydrogen and certain hydrogen carbon compounds which can be removed from coal by heating.

- A. moisture content
- B. product of combustion
- C. ash
- D. volatile matter

Answer:D

By heating 1 gram of coal in an uncovered crucible until the coal is completely burned, the __ will formed.

- A. volatile matter and moisture
- B. ash
- C. Fixed carbon
- D. moisture content

Answer:B

Caking coal are used to produce coke by heating in a coke oven in the absence of __ with volatile matter driven off.

- A. air
- B. oil
- C. oxygen
- D. nitrogen

Answer:A

Grindability of standard coal is

- A. 80
- B.90
- C. 100
- D. 110

Answer:C

Major constituent of all natural gases is

- A. ethane
- B. methane
- C. propane
- D. Cethane

Answer: B

Two types of fans are:

- A. centrifugal and axial
- B. reciprocating and axial
- C. centrifugal and rotary
- D. tangential and rotary

Answer:A

Enthalpy of substance at specified state due to chemical composition.

- A. enthalpy of reaction
- B. enthalpy of combustion
- C. enthalpy of formation
- D. enthalpy of product

Answer:C

A type of boiler used for super critical pressure operation.

- A. La Mont boiler
- B. Once through-circulation boiler
- C. Force circulation boiler
- D. Natural circulation boiler

Answer:B

Economizer in a water tube boiler is heated by:

- A. electric furnace
- B. electric current
- C. incoming flue gas
- D. outgoing flue gas

Answer:D

Receives heat partly by convection and partly by radiation.

- A. radiant superheater
- B. desuperheater
- C. convective superheater
- D. pendant superheater

Answer:D

Regenerative superheater is a storage type of heat exchangers have an energy storage medium called.

- A. matrix
- B. regenerator
- C. Boiler
- D. Recuperator

Answer:A

Stirling cycle uses a _____ as working fluids.

- A. incompressible gas
- B. incompressible fluids
- C. compressible refrigerant
- D. compressible fluids

Answer:D

In Stirling process, the heat is added during

- A. Isobaric process
- B. Isentropic process
- C. Isothermal process
- D. Heat process

Ans. C.

Brayton cycle is known as

- A. Carnot cycle
- B. Joule cycle
- C. Carnot cycle
- D. Rankine cycle

Ans. B

It is applied to propulsion of vehicle because of certain practical characteristics.

- A. Diesel cycle
- B. Otto cycle
- C. Carnot cycle
- D. Brayton cycle

Ans. D

Heat exchangers typically involve

- A. no work interaction
- B. no heat interaction
- C. no energy interaction
- D. none of these

Ans. D

A device that is used to convert the heat to work is called

- A. Adiabatic
- B. Regenerator
- C. Heat engines
- D. None of these

Ans. C

The objective of a heat pump is to maintain a heated space at

- A. Low temperature
- B. High temperature
- C. Medium temperature
- D. None of these

Ans. B

A device that violates the second law of thermodynamics is called

- A. perpetual motion machine of second kind
- B. perpetual motion machine of third kind
- C. perpetual motion machine of the first kind
- D. none of these

Ans. B

A process is called _____ if no irreversibilities occur outside the system boundaries during the process.

- A. externally reversible
- B. internally reversible
- C. reversible
- D. none of these

Ans. A

An energy interaction which is not accompanied by entropy transfer is

- A. energy
- B. heat
- C. work
- D. none of these

Ans. C

A _____ is used in aircraft engines and some automotive engine. In this method, a turbine driven by the exhaust gases is used to provide power to compressor or blower at the inlet.

- A. discharging
- B. turbocharging
- C. supercharging
- D. scavenging

Ans. B

The only device where the changes in kinetic energy are significant are the

- A. compressor
- B. pumps
- C. nozzles and diffusers
- D. none of these

Ans. C

The distance between TDC and BDC in which the piston can travel is the

- A. right extreme position
- B. displacement stroke
- C. stroke of the engine
- D. swept stroke

Ans. C

In compression-engine the combustion of air-fuel mixture is self-ignited as a result of compressing the mixture above its

- A. self developed temperature
- B. mixing temperature
- C. self feed temperature
- D. self ignition temperature

Ans. D

The thermal efficiency of an ideal Otto cycle depends _____ of the working fluid.

- A. the pressure ratio of the engine and the specific ratio
- B. the temperature ratio of the engine and the specific ratio
- C. the moles ratio of the engine and the specific heat ratio
- D. the compression ratio of the engine and the specific heat ratio

Ans. D

Using monoatomic gas, the thermal efficiency of Otto cycle

- A. increases
- B. decreases
- C. remains constant
- D. none of these

Ans. A

In diesel engine, combustion process during combustion occurs during

- A. isothermal process
- B. constant pressure process
- C. isentropic process
- D. adiabatic

Ans. B

If the cutoff ratio decreases, the efficiency of diesel cycle

- A. increases
- B. decreases
- C. remains constant
- D. none of these

Ans. A

If Erickson cycle, the regeneration process occur during _____ process.

- A. constant volume
- B. constant temperature
- C. constant pressure
- D. none of these

Ans. C

In Brayton cycle, the _____ during constant pressure process.

- A. work is added
- B. heat is transferred
- C. pressure is rejected
- D. energy is added

Ans. B

The two major application areas of gas turbine engines are

- A. driving automotive engine and locomotives
- B. heating and generation
- C. aircraft propulsion and electric power generation

D. none of these

Ans. C

The use of regenerator in is recommended only when the turbine exhaust temperature is higher than the compressor.

- A. exit temperature
- B. inlet temperature
- C. mean temperature
- D. absolute temperature

Ans. A.

As the number of stages is increased, the expansion process becomes

- A. isentropic
- B. isothermal
- C. isometric
- D. polytropic

Ans. B

Aircraft gas turbines operate at higher pressure ratio typically between

- A. 6 to 8
- B. 12 to 24
- C. 10 to 18
- D. 10 to 25

Ans: D

The first commercial high-pass ratio engines has a bypass ratio of

- A. 1
- B. 3
- C. 5
- D. 7

Ans: C

The single-stage expansion process of an ideal brayton cycle without regeneration is replace by a multistage expansion process with reheating between the same pressure limits. As a result of modification, thermal efficiency will:

- A. Increase
- B. Decrease
- C. Remain constant
- D. none of these

Ans: B

Which of the following is/are the application of Brayton cycle

- A. Propulsion system
- B. Automotive Turbine Engines
- C. Aircraft Turbine engines
- D. all of these

Ans. D

It used as working fluid in high-temperature application of vapor cycle?

- A. Helium
- B. Deuterium
- C. Mercury
- D. Water

Ans. C

The superheat vapor enters the turbine and expands isentropically and produces work by the rating shaft. The _____ may drop, during the process.

- A. density
- B. Viscosity of fuel
- C. Temperature and pressure
- D. none of these

Ans. C

Only _____ of the turbine work output is required to operate the pump

- A. 0.01%
- B. 0.02%
- C. 0.03%
- D. 0.04%

Ans. D

Superheating the steam to higher temperature decreases the moisture content of the steam at the _____

- A. turbine inlet
- B. compression inlet
- C. compressor exit
- D. turbine exit

Ans. D

Regeneration also provides a convenient means a dearating the feedwater to prevent

- A. boiler explosion
- B. boiler scale production
- C. boiler corrosion
- D. compressor damage

Ans. C

Can be apply in Steam turbine cycle (Rankine), gas turbine cycle (Brayton) and combined cycle

- A. Hydroelectric plant
- B. Nuclear power plant
- C. Cogeneration plant
- D. Tidal power plant

Ans. C

In a Rankine cycle with fixed turbine inlet conditions. What is the effect of lowering the condenser pressure the heat rejected will.:

- A. increase
- B. decrease
- C. remains the same
- D. none of these

Ans. B

In an ideal Rankine cycle with fixed boiler and condenser pressures. What is the effect of superheating the steam to a higher temperature, the pump work input will:

- A. increase
- B. decrease
- C. remains the same
- D. none of these

Ans. C

How do the following quantities change when the simple ideal Rankine cycle is modified with? The heat rejected:

- A. increase
- B. decrease
- C. remains the same
- D. none of these

Ans. B

During a combustion process, the components which exist before the reaction are called

- A. reaction
- B. combustion
- C. reactants
- D. product

Ans. C

In an obvious reason for incomplete combustion

- A. insufficient carbon
- B. insufficient air
- C. insufficient nitrogen
- D. insufficient oxygen

Ans. D

Higher heating value when H₂O in the product of combustion is in

- A. solid form
- B. vapor form
- C. gas form
- D. liquid form

Ans. D

Device which transfer heat from low temperature medium to a higher temperature one is a

- A. adiabatic
- B. refrigerator
- C. heat exchanger
- D. heat pump

Ans. D

A rule of thumb is that the COP improves by _____ for each C the evaporating temperature is raised or the condensing temperature is lowered.

- A. 2 to 4%
- B. 6 to 7%
- C. 1 to 5%
- D. 6 to 10%

Ans. A

Are generally more expensive to purchase and install than other heating systems , but they save money in the long run.

- A. Refrigerator
- B. Adiabatic
- C. Heat pumps
- D. Humidifyer

Ans. C

The most widely used absorption system is the ammonia-water system, where ammonia is serves as a refrigerant and H₂O as the

- A. cooling
- B. heating
- C. heating and cooling
- D. transport medium

Ans. D

The efficiency of all reversible heat engines operating between the same two reservoir _____

- A. Differ
- B. Are the same
- C. Are Unequal
- D. None of the above

Ans. B

A process with no heat transfer is known as

- A. isobaric process
- B. adiabatic process
- C. isothermal process
- D. isothermal process

Ans. B

The relative density of a substance is the ratio of its density of:

- A. mercury
 - B. oil
 - C. gas
 - D. water
- Ans. D

This type of heat exchanger allows fluids to flow at right angles to each other

- A. Series flow
 - B. Parallel flow
 - C. Cross flow
 - D. Counter flow
- Ans. C

The fact the total energy in any one energy system remains constant is called the principle of _____

- A. Conservation of Energy
 - B. Second Law of Thermodynamics
 - C. Conservation of Mass
 - D. Zeroth Law of Thermodynamics
- Ans a

A process for which the inlet and outlet enthalpies are the same

- A. isenthalpic
 - B. enthalpy conservation
 - C. throttling
 - D. steady state
- ans c

the sum of energies of all the molecules in system, energies that appear in several complex forms

- a. kinetic energy
 - b. internal energy
 - c. external energy
 - d. flow work
- ans B

a system that is completely impervious to its surrounding. Neither mass nor energy cross its boundaries

- a. open system
 - b. closed system
 - c. adiabatic system
 - d. isolated system
- ans d

a device used to measure small and moderate pressure difference

- a. manometer
 - b. bourdon gage
 - c. barometer
 - d. piezometer
- ans a

a vapor having a temperature higher than the saturation temperature corresponding to its pressure.

- a. superheated vapor
 - b. saturated vapor
 - c. super saturated vapor
 - d. subcooled vapor
- ans a

the energy or stored capacity for performing work possessed by a moving body, by virtue of its momentum.

- a. internal energy
- b. work
- c. gravitational potential energy

- d. kinetic energy
- ans D

the thermodynamic process wherein temperature is constant and the change in internal energy is zero

- a. isobaric process
 - b. isometric process
 - c. isothermal process
 - d. polytropic process
- ans c

the function of a pump and compressor is to

- a. transfer heat from one fluid to another
 - b. increase the total energy content of the flow
 - c. extract energy from the flow
 - d. exchange heat to increase energy to the flow
- ans b

this law states that 'all energy received as heat by a heat-engine cannot be converted into mechanical work'

- a. 1st law of thermodynamics
 - b. 2nd law of thermodynamics
 - c. 3rd law of thermodynamic
 - d. all of the above
- ans b

the intensity of pressure that is measured above absolute zero is called

- a. gage pressure
 - b. absolute pressure
 - c. vacuum pressure
 - d. saturation pressure
- ans b

this is the ratio of the heat equivalent of the brake or useful horsepower developed by an engine and available on its crankshaft to the heat during the same time

- a. brake engine efficiency
 - b. indicated thermal efficiency
 - c. combined thermal efficiency
 - d. brake thermal efficiency
- ans d

flow work is equal to pressure times _____

- a. temperature
 - b. entropy
 - c. internal energy
 - d. specific volume
- ans d

this form of energy is due to the position or elevation of the body

- a. internal energy
 - b. kinetic energy
 - c. potential energy
 - d. work
- ans c

another term for constant volume process

- a. isometric
 - b. isochoric
 - c. isovolumic
 - d. all of the above
- ans D

work done by the steam during a reversible adiabatic expansion process in the turbine

- a. brake work
 - b. ideal work
 - c. actual fluid work
 - d. combine work
- ans b

the efficiency of carnot cycle depends upon the

- a. pressure
 - b. entropy
 - c. volume
 - d. temperature
- ans D

is the heat required in a constant pressure process to completely vaporiz a unit-mass of liquid at a given temperature

- b. latent heat vaporization
 - c. enthalpy of vaporization
 - d. all of the above
- ans d

it is a commonly used device for measuring temperature differences or high temperatures.

- a. thermistor
 - b. thermocouple
 - c. bimetallic strip
 - d. mercury in glass
- ans b

the science and technology concerned with precisely measuring energy and enthalpy

- a. thermodynamics
 - b. chemistry
 - c. calorimetry
 - d. none of the above
- ans c

the rate of doing work per unit time

- a. torque
 - b. power
 - c. force
 - d. moment
- ans B

if an ideal rankine cycle with fixed boiler and condenser pressure. What is the effect of superheating the steam to a higher temperature to the cycle thermal efficiency?

- a. the cycle thermal efficiency will increase
 - b. the cycle thermal efficiency will decrease
 - c. the cycle thermal efficiency will remain constant
 - d. none of the above
- ans a

a vapor having a temperature higher than the saturation temperature corresponding to the existing pressure

- a. superheated vapor
 - b. saturated vapor
 - c. wet vapor
 - d. none of the above
- ans a

it is the work done in pushing a fluid across a boundary, usually into or out a system

- a. mechanical work

- b. non flow work
 - c. flow work
 - d. electrical work
- ans c

a liquid that has a temperature lower than the saturation temperature corresponding to the existing pressure.

- a. subcooled liquid
 - b. saturated liquid
 - c. unsaturated liquid
 - d. water
- ans a

this type of boiler, the water passes through the tubes while the flue gases burn outside the tubes

- a. water column
 - b. try cocks
 - c. gauge glass
 - d. all of the above
- ans c

it prevents damage to the boiler by giving warning of low water

- a. safety valve
 - b. fusible plug
 - c. relief valve
 - d. try cocks
- ans b

it has several functions. When necessary it empties the boiler for cleaning, inspection, or repair. It blows out mud scale, or sediment when the boiler is in operation and prevents excessive concentration of soluble impurities in the boiler

- a. blow-down line
 - b. boiler feedwater pump
 - c. steam valve
 - d. none of the above
- ans a

is a feedwater preheating and waste heat recovery device which utilizes the heat of the flue gases

- a. economizer
 - b. open heater
 - c. closed heater
 - d. waterwalls
- ans a

it is a heat exchanger which utilizes the heat of the flue gases to preheat the air needed for combustion

- a. economizer
 - b. feedwater heater
 - c. reheater
 - d. air preheater
- ans d

it is a sustem of furnace cooling tubes which can extend the evaporative capacity of the water-tube boiler and at the same time protect the furncace walls from high temperature.

- a. reheater
 - b. waterwalls
 - c. superheater
 - d. feedwater heater
- ans b

it is based on the generation of 34.5 lbm/hr of steam from water at 212F and equivalent to 33500 btu/hr

- a. one hp
 - b. one kw
 - c. one Boiler Hp
 - d. none of the above
- ans c

it prevents boiler pressure from rising above a certain predetermined pressure by opening to allow excess steam to escape into the atmosphere when that point is reached, thus guarding against a possible expulsion through excessive pressure.

- a. relief valve
 - b. safety valve
 - c. fusible plug
 - d. pressure switches
- ans B

in a water tube boiler, the water will pass through _____

- a. inside the tubes
 - b. outside the tubes
 - c. inside the shell
 - d. outside the shell
- ans a

it is the temperature to which the air becomes saturated at constant pressure

- a. dry bulb temperatue
 - b. wet bulb temperature
 - c. dew point temperature
 - d. saturation temperature
- ans C

in a _____ cooling tower, the air moves horizontally through the fills as the water moves downward.

- a. cross flow
 - b. counter flow
 - c. parallel flow
 - d. double flow
- ans a

it is the subject that deals with the behavior of moist air

- a. psychrometer
 - b. psychrometry
 - c. refrigeration
 - d. pneumatics
- ans b

it is the ration of the mass of water vapor in a certain volume of moist air at a given temperature to the mass of hot water in the same volume of saturated air at the same temperature

- a. humidity ratio
 - b. specific humidity
 - c. humidity
 - d. relative humidity
- ans D

air whose condition is such that any decrease in temperature will result in condensation of water vapor into liquid.

- a. saturated air
 - b. unsaturated air
 - c.saturated vapor
 - d. moist air
- ans a

it is the warm water temperature minus the cold water temperature leaving the cooling tower

- a. approach

- b. terminal difference
 - c. cooling range
 - d. LMTD
- ans c

the temperature where the relative humidity becomes 100 % and where the water vapor starts to condense is known as _____

- a. dry bulb temperature
 - b. dewpoint temperature
 - c. wet bulb temperature
 - d. saturated temperature
- ans b

the surrounding air _____ temperature is the lowest temperature to which water could possibly be cooled in a cooling tower

- a. dry bulb
 - b. wet bulb
 - c. dew point
 - d. saturation temperature
- ans b

which is not a major part of the vapor compression system?

- a. compressor
 - b. condenser
 - c. evaporator
 - d. refrigerant
- ans d

this refers to the rate of heat transfer attribute only to a change in dry-bulb temperature

- a. sensible heating or cooling
 - b. humidification
 - c. dehumidification
 - d. cooling and dehumidifying
- ans a

it is a binary mixture of dry air and water-vapor

- a. dry air
 - b. saturated vapor
 - c. moist air
 - d. wet mixture
- ans c

the temperature measured by an ordinary thermometer

- a. wet bulb temp
 - b. dry bulb temp
 - c. dew point temp
 - d. wet – bulb depression
- ans b

the mass of water interspersed in each kilogram of dry air

- a. enthalpy
 - b. humidity ratio
 - c. specific volume
 - d. relative humidity
- ans b

this system combines two vapor compression units with the condenser of the low temperature system discharge its heat to the evaporator of the high temperature system

- a. cascade systems
- b.multistage system
- c. binary system
- d. multi pressure system

ans a

a process of increasing the humidity ratio at constant dry bulb temperature

- a. dehumidifying process
- b. cooling process
- c. heating process
- d. humidifying process

ans d

the ratio of the partial pressure of water vapor in the air to the saturation pressure corresponding to the temperature of the air

- a. humidity ratio
- b. relative humidity
- c. specific humidity
- d. moisture content

ans b

in an air conditioning process that involves heating without changing the moisture content of air. The process is represented by a horizontal line in the psychrometric chart, from left to right

- a. sensible cooling process
- b. sensible heating process
- c. humidifying process
- d. heating and dehumidifying process

ans b

it is an air conditioning process of increasing the humidity ratio without changing the dry-bulb temperature of air. The process is represented in the psychrometric chart by a vertical line, from up to down

- a. sensible cooling process
- b. sensible heating process
- c. humidifying process
- d. heating and dehumidifying process

ans C

the temperature at which the water vapor content of moist air begins to condense when air is cooled at constant pressure

- a. dew point temp
- b. wet bulb temp
- c. dry bulb temp
- d. condensing temp

ans a

it is the index of performance of a refrigeration system which is a dimensionless quantity

- a. coefficient of performance
- b. energy ratio
- c. energy efficiency
- d. performance ratio

ans a

it is simply the compression of the gas in two or more cylinders in place of a single cylinder compressor

- a. intercooled compression
- b. multistage compression
- c. efficient compression
- d. performance compression

ans b

the transfer of energy from the more energetic in two or more energetic particles of a substance to the adjacent less energetic ones as a result of interaction between the particles

- a. heat transfer
- b. radiation

c. conduction

d. convection

ans c

what is the simultaneous control of temperature, humidity, air movement, and quantity of air in space?

- a. refrigeration
- b. psychrometry
- c. air conditioning
- d. humidification

ans c

the non condensing component of the moist air

- a. hydrogen
- b. water vapour
- c. nitrogen
- d. dry air

ans d

the substance used for heat transfer in a vapor compression refrigerating system. It picks up heat by evaporating at a low temperature and pressure and gives up this heat by condensing at a higher temperature and pressure

- a. water
- b. air
- c. ammonia
- d. gas

what is the pressure of the refrigerant between the expansion valve and the intake of the compressor in a multi pressure refrigeration system?

- a. high side pressure
- b. discharge pressure
- c. condensing pressure
- d. low side pressure

ans D